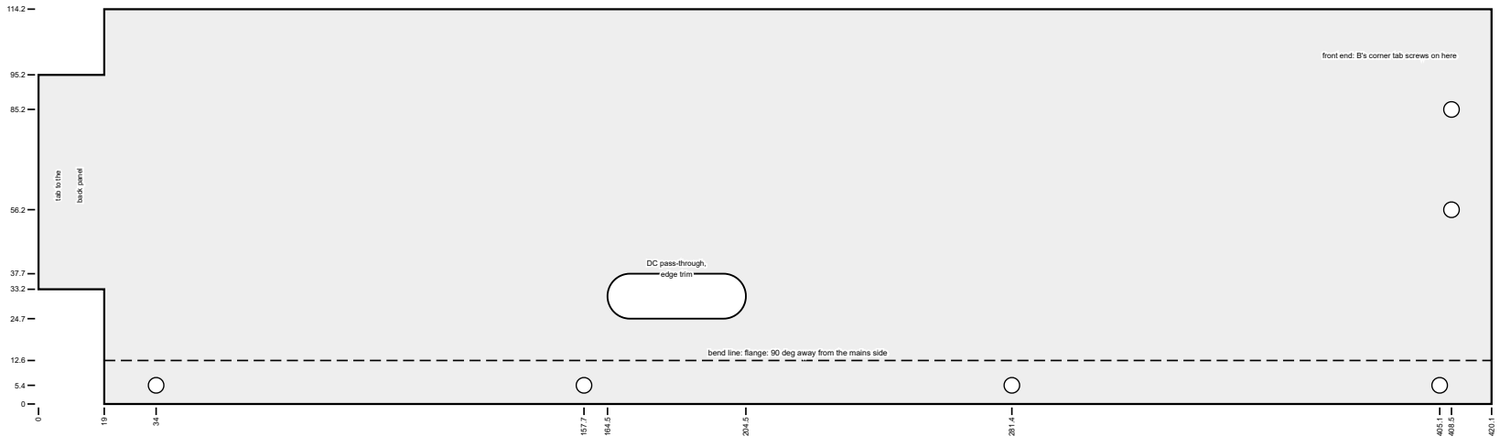
RMR control panel: mains compartment wall

Two pieces of 1/16 in 5052 that close the 110 VAC compartment on the lower deck (sheet 3, view B), cut from the offcut of the upper plate's sheet. Flat patterns and section 1:2, plan 1:5, printed at 100 %. Millimetres.

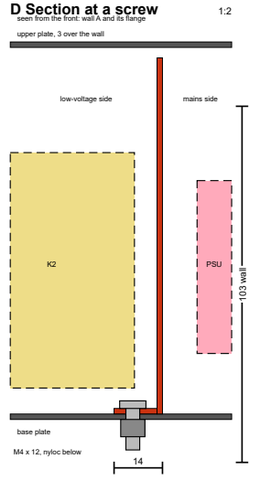
A Wall A, flat, seen from the mains side

420.1 x 114.2; along x, mains face at y 182



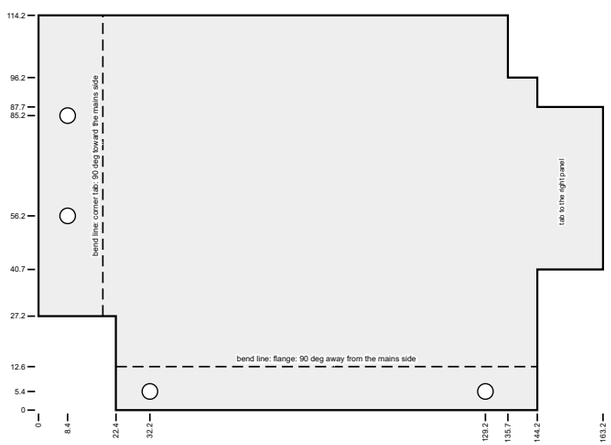
D Section at a screw

1:2



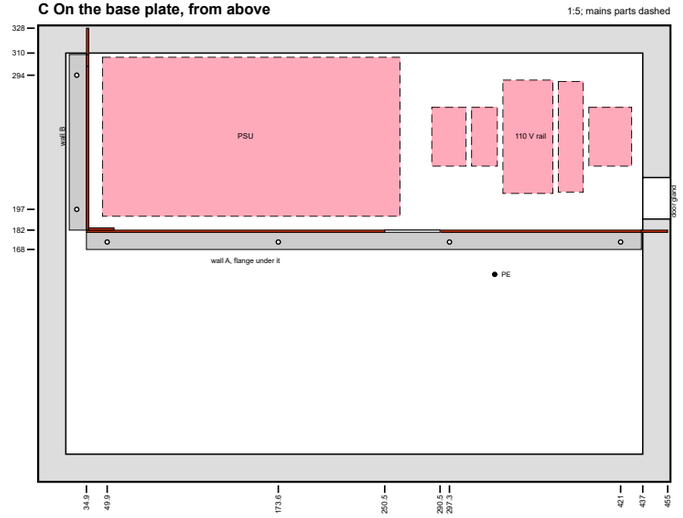
B Wall B, flat, mains side

163.2 x 114.2; along y



C On the base plate, from above

1:5; mains parts dashed



Material and cut

1/16 in 5052-H32 from the offcut of aluminum 2, beside the upper plate (sheet 4): flat A 420.1 x 114.2, B 163.2 x 114.2. Shear the blanks, nibble the end tabs and B's corner relief, drill the Ø4.5 holes, cut the pass-through, deburr; DXF: wall_mains_flat.dxf (layers CUT and BEND).

Bends

90 deg on a brake, inside radius one thickness. K 0.42 is assumed: bend deduction 2.8 per bend. Bend a test strip first and correct WallK if its legs come out long or short. Flanges fold away from the mains side; B's corner tab folds toward it and lies on A's mains face.

Fastening and PE

Flanges: 6 M4 x 12 into the base plate, nyloc below, 3.8 mm past the nut (NutFits). Corner: 2 M4 x 12 through B's tab and A, heads on the mains side. Drill the base plate through the flange holes in place. Each piece is its own PE part: a ring lug under a flange screw.

Openings

End tabs reach between the rings to 2 mm from the back and right panels; A and B stop 1 mm short of the rings. B's tab, and a notch in the top of B's right end, keep 1 mm from the front right corner brackets and their screw heads, taken as 20 mm legs (BracketChecks). Measure a bracket before cutting B: a 4 mm key must reach both screws of the front right post to mid ring bracket past B's tab, its post screw along x and its rail screw from below; if not, lower the tab's top (TabHB) until it does. The top stops 3 mm under the upper plate. DC pass-through 40 x 13, 20 over the deck, centred on the mains duct: PSU V+ and V- out. Move it to the PSU's terminal end once that face is known. The 110 V door cable leaves by an M20 gland in the back panel (u 205), not through the wall.

Upper plate

Drill no DIN rail or duct screw in these bands: its nyloc and tail hang 9.4 mm under the plate, and the wall stops 3 mm under it.
A: x 29.9 to 441, y 175.4 to 187
B: x 29.9 to 41.5, y 177 to 305.5
B corner tab: x 31.5 to 60, y 177 to 188.6

Fit

Before the lower deck's parts: flange screws, then B's corner tab to A, then the PE jumpers. Check that the upper module drops on clear.

Mains compartment wall

Sheet 5. Legal, landscape, 100 %.
Flat patterns and section 1:2, plan 1:5.
Designed 2026-10-18 (issue #146).
DXF: wall_mains_flat.dxf.